Getting started with mqTrains Relay 16 Board

mqTrains OVERVIEW

ESP8266 microprocessors are an inexpensive option for controlling model railroad accessories such as turnouts, sensors and signals. They connect quite easily to JMRI using the ESP's WiFi interface and the MQTT messaging protocol. ESP01s can be purchased from online stores for about a couple of bucks each.

mqTrains sketches have been developed or are in development to handle some common model train requirements such as turnout control, signalling and occupancy detection. Visit mqTrains.com for more information.

ESP devices only use the 2.4 GHz WiFi band. Some common domestic WiFi routers are limited to 16 or 32 concurrent sessions. Check your router specifications if you intend to have a high number of these devices.

mqTrains Relay 16 Board

The mqTrainsRelay16 sketch runs on the "ESP8266 WIFI 16 Channel Relay Module ESP-12F Development Board" (DC 5V, 7-28V or 12V supply depending on the version). The sixteen relays are driven through the board's 74HC595 shift registers (data GPIO14, latch GPIO12, clock GPIO13, enable GPIO5), and all sixteen are switched over MQTT on items `/01` to `/16`.

LET'S GET STARTED

These are the main steps to get started with this mqTrains board. More complex features are covered in the mqTrains User Reference.

1. Install the MQTT Server Software

Linux, Mac and Pi users:

We recommend using a Raspberry Pi for the MQTT Server Software. The Pi Zero W is quite adequate and inexpensive. Follow the vendor instructions to set up the Raspberry Pi. To install Mosquitto Server, use the commands:

```
sudo apt-get update
sudo apt-get install mosquitto mosquitto-clien
ts
```

Mosquitto version 2 (since December 2020) uses tighter security rules. Create this config file:

```
sudo nano /etc/mosquitto/conf.d/mosquitto.conf
```

Insert these two lines and then press ctrl-o <enter> and ctrl-x

```
listener 1883
allow_anonymous true
```

and restart mosquitto with

```
sudo service mosquitto restart
```

Windows users:

Mosquitto can also be installed on most Windows computers. Refer to the Mosquitto website for details. If you install on Windows, you will most likely need to grant Firewall permission. Instructions for granting permission with the Windows Defender Firewall are shown at the end of this document. If you have a Firewall with non-Microsoft security software, refer to the vendor instructions.

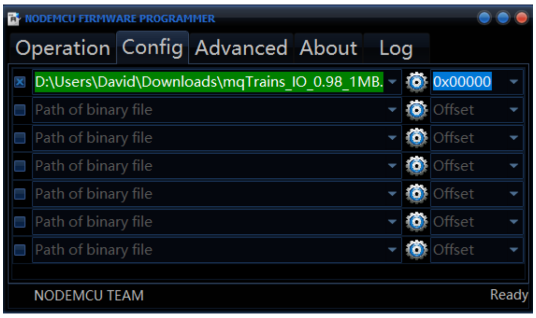
2. Install mqTrains on your ESP-12F relay board

Download the mqTrains binary file from here:
mqTrains.com/downloads/

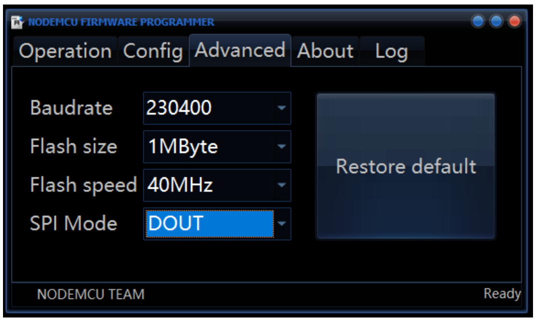
Windows users: Download the nodemcu-flasher program from the appropriate link: x64 version (most common) or x32 version (for old 32-bit computers).

The relay board has an on-board ESP-12F module and a programming header instead of a removable ESP01. Connect a 3.3 V USB-serial adapter to the header: adapter TX to board RX, adapter RX to board TX and GND to GND. To put the ESP8266 in flash mode, hold GPIO0 (IO0) to GND while powering the board up, then release it. After flashing, power-cycle the board without IO0 grounded.

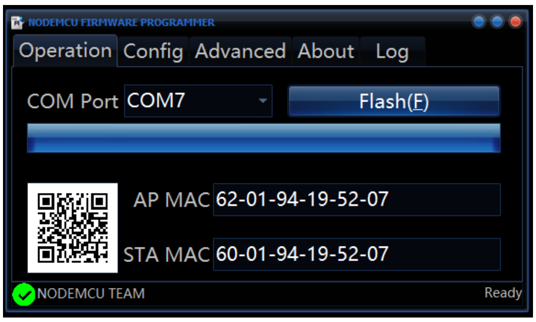
Run the installer (double click the exe file downloaded), select the Config tab, click on the wheel in the first row, then navigate to and select the binary file you downloaded.



Select the Advanced tab and select DOUT for SPI Mode:



On the Operation tab, the correct COM Port will most likely be selected for you, change if need be, then click on Flash to start the upload. The blue bar will show the upload progress which takes about 3 minutes. Wait for the green tick and "Ready" message at the bottom of the screen.



Linux, Mac and Pi users:

```
sudo apt-get update
sudo apt-get install python python-pip
sudo pip install esptool
```

Then, while in the folder with the .bin file, run (all in one line and with the correct serial port, ttyUSB0 shown here):

```
esptool.py --chip esp8266 --port /dev/ttyUSB0
--baud 460800 --before default_reset --after h
ard_reset write_flash 0x0 mqTrains_Relay16_x.x
x_4MB.bin
```

Note: this board uses a 4 MByte flash, so in the nodemcu-flasher Advanced tab select 4MByte as Flash size (and DOUT SPI mode as shown).

3. Wiring your relay board

Power the board per the markings on your version (5 Vdc regulated, or the wide-range DC input). Each relay provides COM, NO and NC screw terminals; wire your accessories to COM+NO (normally open) or COM+NC (normally closed) as required, keeping the load within the relay rating.

4. Configuring your relay board

Using a smartphone or computer, look in your available WiFi networks list for the mqTrains Access Point (AP) network service started by the ESP. The default SSID name is “**mqTrains-9001**” with the Key or password set to “**mqtrains**”.

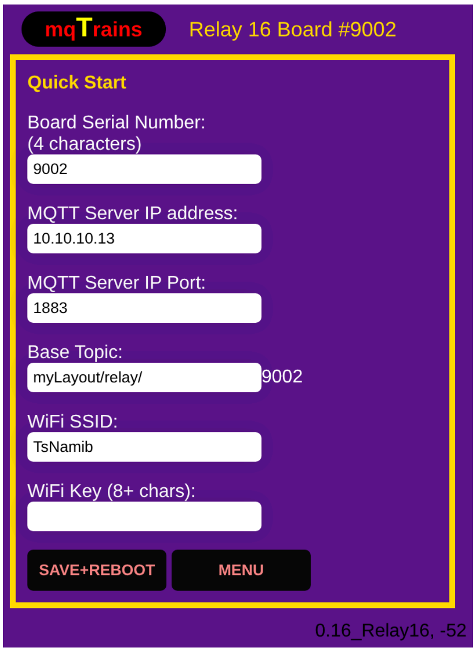
Once connected to the ESP’s AP, use a web browser to connect to the mqTrains configuration page using http://2.2.2.1.

You can use these QR codes to connect to the default Access Point and web page. (The 2nd QR code for the web page won’t work if your app needs the internet to decode, like Google Lens.)


WiFi mqTrains-9001


Web Page 2.2.2.1

Select the Quick Start option from the main menu and change each of these values to suit your needs.



Set a 4 digit Serial Number for the board. If you have more than one mqTrains board, each should have a unique serial number. This will be used as part of the MQTT messaging. We recommend not leaving the default value, since the next new board will also be 9001. Some use 9000s for relay boards; others use two numbers for a location on the layout and two more for a serial number at that place.

Set the IP address of the computer on which your MQTT Server (Broker) is installed and the port number it uses. 1883 is the default port.

Set the base portion of the MQTT topic. The full topic will have the board serial number and the two digit item number appended to the base. For example, using the default base topic, the full MQTT topic for relay number /03 on board number 9002 will be `myLayout/relay/9002/03`.

Set the WiFi SSID and Key to connect to your desired WiFi service. Keep in mind that the WiFi service will need to provide you access to the MQTT Server whose IP address you entered earlier.

Press SAVE+REBOOT to save all these values to the ESP's internal storage and restart it using those values.

Using a utility such as MQTT Explorer or an MQTT subscribe line command, look for messages for topics beginning with the Base Topic. Expect to see a message like "192.168.1.10 is here!" This is the IP Address of the ESP on your WiFi network. Use this IP address in your browser to connect to the configuration page to complete the set up or to make changes. The AP will no longer be needed and will shut down once the ESP is able to post a message to the MQTT Server.

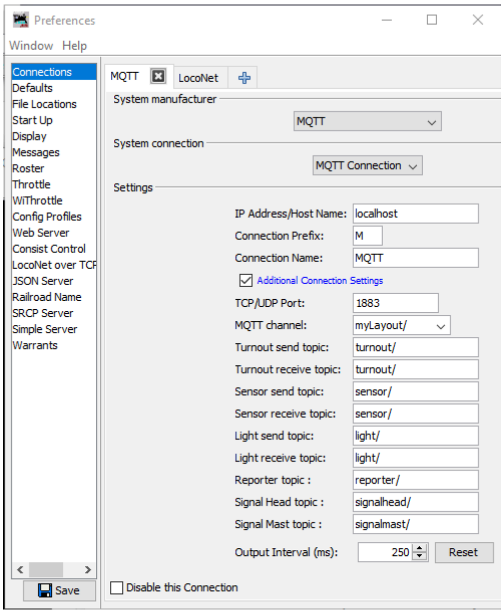
Prefer a serial cable? These boards can also be provisioned over the serial port (115200 baud): send `:h;` for the command list, then for example `:Bserial 9001;`, `:Bbase myLayout/relay/;` and `:Bsskr0000 MySSID MyWiFiKey;` followed by a reboot.

5. Configure the relays

Select *Pin detail* from the main menu. For each relay, the *Enabled* box must be ticked for relays being used and should be unchecked for unused relays — no messages are processed for disabled relays. *Locked* makes a relay ignore commands temporarily.

Linqs let relays move together: give two or more relays the same Linq number (1-8) and they will follow each other; tick *Invert Linq* on a relay that must do the opposite of its group. This is handy for crossovers or for frog polarity with a second relay.

Relay states are monitored using the Live View page. You can also toggle a relay by tapping on its state box, and move a whole linq group by tapping its L badge. Each change publishes an MQTT message so JMRI stays in step.



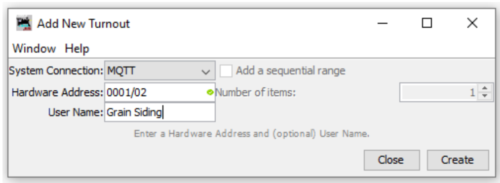
JMRI Connections (v 4.21.2)

Each relay listens for the payloads `ON`, `THROWN`, `ACTIVE` or `1` to energise and `OFF`, `CLOSED`, `INACTIVE` or `0` to release; `TOGGLE` (or `2`) flips the state. The board publishes `ON`/`OFF` by default; this can be changed on the MQTT Info page.

6. Set JMRI to talk with your ESP01

In JMRI, using Edit -> Preferences -> Connections, create an MQTT connection with the settings as shown, and especially make sure you specify the same host on which you have installed the MQTT Server as you provided to the ESP. You can specify the host name, or the IP address as you did on the ESP01. If the MQTT server is running on the same computer as JMRI, you can also use `localhost` as shown.

Create MQTT lights (for ON/OFF messaging) or turnouts using the board serial number and the 2 digit relay number for the hardware address, for example the third relay on board 9002 would have a hardware address of `9002/03`.



JMRI Add New Turnout

Try mqTrains from the command line

The mqTrains web pages allow you to see and change relay states. On a command line, with the mosquitto tools installed, you can see the messages being published while mqTrains is connected to the MQTT Server:

```
   prompt> mosquitto_sub -v -h localhost -t "myLayout/#"
or in Windows
   "c:\Program Files\mosquitto\mosquitto_sub.exe" -v -h localhost -t "myLayout/#"

   -v means verbose - show the full message (topic and payload).
   -t means topic
   -h means the MQTT Server hostname or IP address
   -r means retain the message in the Server.
   -m means message, aka payload.
```

You can publish messages (ON or OFF) to the MQTT Server and see the relay switch and the state change on the web page or in JMRI:

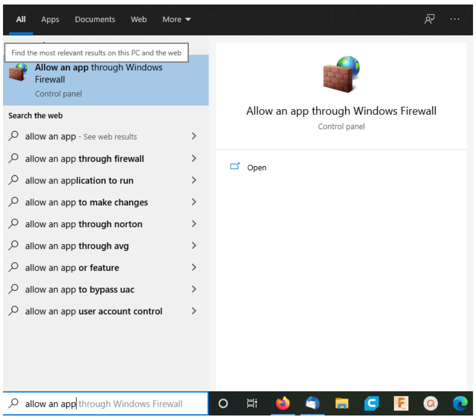
```
   prompt> mosquitto_pub -h localhost -r -t "myLayout/relay/9002/03" -m "ON"
   prompt> mosquitto_pub -h localhost -r -t "myLayout/relay/9002/03" -m "OFF"
   prompt> mosquitto_pub -h localhost -t "myLayout/relay/9002/03" -m "TOGGLE"
```

Note that `xxxx/00` is used for board specific things (see the mqTrains-UserReference.pdf for more information) and the relays are from `xxxx/01` to `xxxx/16`.

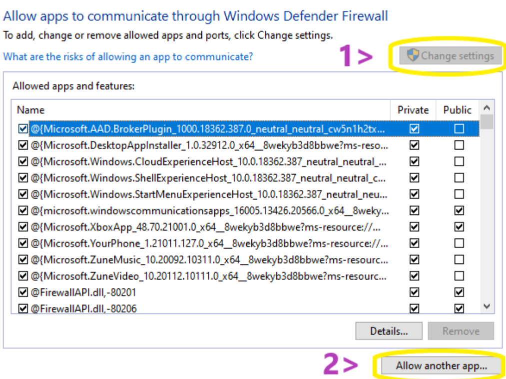
Burning through the Windows Firewall

If you install MQTT Server Software (mosquitto) on a Windows computer, such as your JMRI machine, you may find that the ESP nodes cannot connect to the MQTT Server due to the Windows Defender Firewall blocking WiFi traffic coming into the computer. This section describes how to alter the Firewall to allow MQTT messages to pass through.

Type "allow an app" into the Windows search bar. The "Allow an app through the Windows Firewall" will appear.

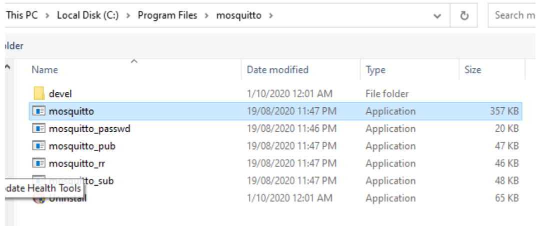


Click on this and then, on "Change Settings" and "Allow another app".

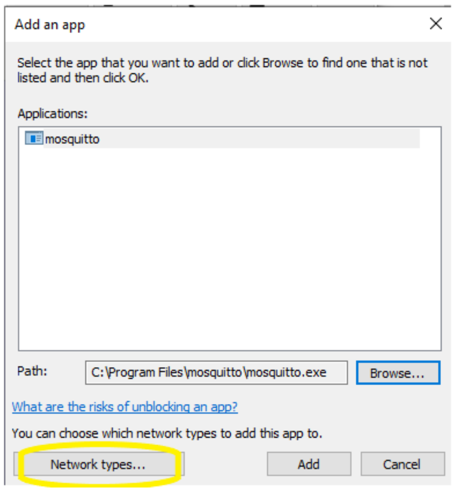


With Browse, select mosquitto.exe from C:\Program Files\mosquitto

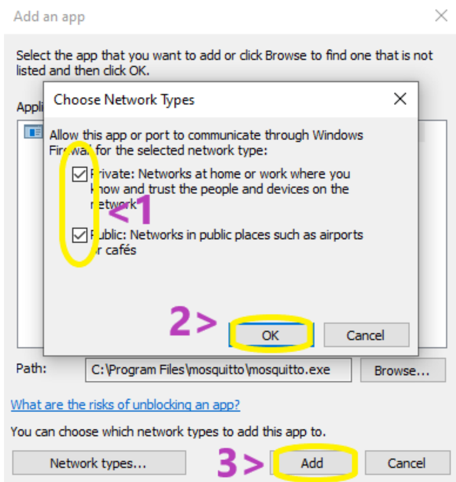
(32-bit machines would have X86 in that folder name)



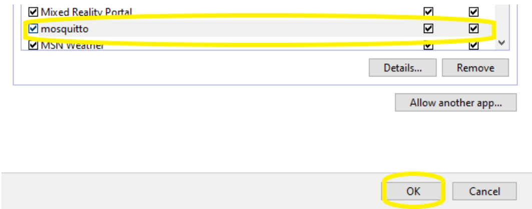
And click on "Open", then click on "Network types".



Depending on how your WiFi network is configured, you may require both Private and Public options so enable both anyway. Then click on "OK" and "Add".



You will then see mosquitto added to the list. Now press OK to finish.



Credits

mqTrains uses, without modifying any of their respective code as available on github:

ArduinoJSON by Bernoit Blanchon
Arduino Queue by Einar Arnason
PubSubClient by Nick O'Leary

LittleFS file system from earlephilhowe
ESPAsyncTCP and ESPAsyncWebServer from me-no-dev